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V300 ZEFIRO

A New Sense of Speed

Catering-VF.CONCEPT SFRS

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Content

1 Introduction 4

1.1 Train Configuration 4

1.2 Definitions and abbreviations 4

2 System Overview 4

2.1 System Configuration 4

3 Functional requirements 5

3.1 Function Fresh Water supply supervision 5
3.1.1 Function Design 5
3.1.2 Requirements 5

3.2 Function Retention tank supervision 6
3.2.1 Function Design 6
3.2.2 Requirements 7

3.3 Function Heating of Water Pipes 7
3.3.1 Function Design 7
3.3.2 Requirements 7

3.4 Function Heating of waste water tanks 8
3.4.1 Function Design 8
3.4.2 Requirements 8

3.5 Function Freeze Drainage 9
3.5.1 Function Design 9
3.5.2 Requirements 9

4 Signal Interface 10

4.1 I/O Interface 10
4.1.1 Supplier system 10
4.1.2 TCMS SW 10

4.2 Bus Interface 11

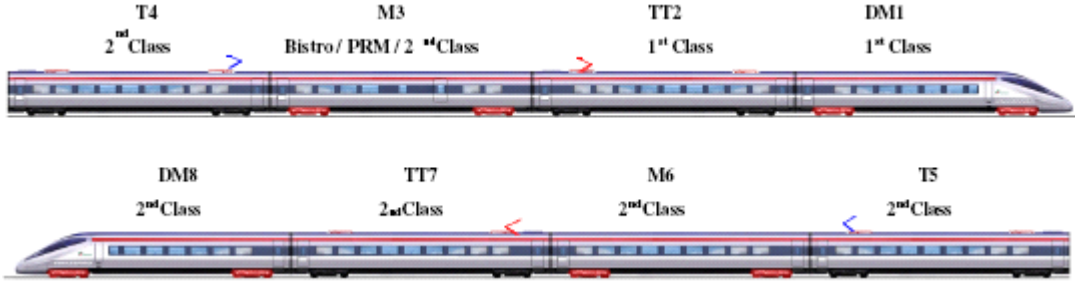
5 References 11

1 Introduction

This document is a function specification for the control of the *Bistrot* system for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



Picture 1 -- Configuration of the system

- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
TCMS	Train Control and Management System
FWT	Fresh Water Tank
WWT	Waste Water Tank

2 System Overview

2.1 System Configuration

The bistrot system is fitted on the coach M3 only.

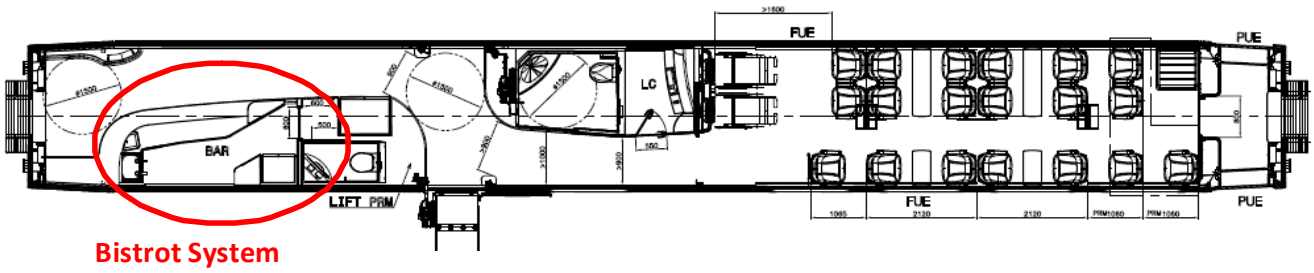


Figure 1 – Bistrot system location

All the functionalities of the bistrot system (refrigerators, glass washer, coffee machine, waste tank heating, boiler, ovens and so on) are completely handled by the bitrot system itself.
There is not bus communication between the TCMS and the Bistrot system.
The data exchange with the TCMS, limited to the current levels of the fresh water tank (FWT) and waste water tank (WWT), is realized by digital I/O signals.

3 Functional requirements

3.1 Function Fresh Water supply supervision

3.1.1 Function Design

The actual level of the FWT is sent to the TCMS via digital I/O for supervision purpose.
The FWT levels transmitted to the TCMS are 0%, 25% (only provisions), 50% (only provisions), 75% (only provisions) and 100% (only provisions). They are shown on a suitable IDU page of the cabin crew (only provisions).
TCMS shall store an event (Warning) when the FWT is completely empty (FWT level 0%).

3.1.1.1 Normal condition

Continuous monitoring of the FWT level (only provisions).

Continuous monitoring of the FWT level 0% status.

3.1.1.2 Failure cases

a) Level transducer failure. Loss of FWT levels indication on the cabin crew IDU (only provisions)
Level switch failure.

3.1.1.3 Failure detection

a) Failure condition detected by the FWT level control unit (only provisions).
No automatic diagnostic function present.

3.1.1.4 Failure handling

a) No action for the driver/crew. Level transducer has to be changed during maintenance.
No action for the driver/crew. Level switch has to be changed during maintenance.

3.1.2 Requirements

ID	2F_08.03.Zefiro-Europe.TRS.11
Description	Fresh Water supply in the bistro area shall be supervised. Information on the level in the fresh water tank shall be shown on the IDU (only provisions): · FWT level 0% (only provisions) · FWT level 25% (only provisions) · FWT level 50% (only provisions) · FWT level 75% (only provisions) · FWT level 100% (only provisions)
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.03.Zefiro-Europe.CONCEPT.27	The following FWT levels shall be transmitted to the TCMS: · FWT level 0% · FWT level 25% (only provisions) · FWT level 50% (only provisions) · FWT level 75% (only provisions) · FWT level 100% (only provisions)	Derived Requirement	4S_08.04.-.Catering		N/A
2F_08.03.Zefiro-Europe.CONCEPT.74	The FWT levels shall be sent to the TCMS via dedicated hardwired lines (one for each level).	Derived Requirement	4S_08.04.-.Catering		N/A
2F_08.03.Zefiro-Europe.CONCEPT.75	The FWT levels shall be transmitted to the TCMS via dedicated hardwired lines (one for each level).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_08.03.Zefiro-Europe.CONCEPT.76	The TCMS shall receive the actual level of the FWT and show them on a suitable IDU page (only provisions).	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_08.03.Zefiro-Europe.CONCEPT.77	TCMS shall store an event (Warning) when the FWT is completely empty (FWT level 0%).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.2 Function Retention tank supervision

3.2.1 Function Design

The actual level of the WWT is sent to the TCMS via digital I/O for supervision purpose.
The WWT levels transmitted to the TCMS are 50% (only provisions), 75% (only provisions) and 100%. They are shown on a suitable IDU page (only provisions).
TCMS shall store an event (Warning) when the WWT is completely full (WWT level 100%).

3.2.1.1.1 Normal condition

Continuous monitoring of the WWT level (only provisions).

Continuous monitoring of the WWT level 100% status.

3.2.1.1.2 Failure cases

a) Level transducer failure. Loss of WWT levels indication on the cabin crew IDU (only provisions).
Level switch failure.

3.2.1.1.3 Failure detection

a) Failure condition detected by the WWT level control unit (only provisions).
No automatic diagnostic function present.

3.2.1.1.4 Failure handling

No action for the driver/crew. Level transducer has to be changed during maintenance.
No action for the driver/crew. Level switch has to be changed during maintenance.

3.2.2 Requirements

ID	2F_08.03.Zefiro-Europe.TRS.13
Description	Retention tank in the bistro area shall be supervised. Information on the level in the retention tank shall be shown on the IDU (only provisions): · WWT level 50% (only provisions) - WWT level 75% (only provisions) - WWT level 100% (only provisions)
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.03.Zefiro-Europe.CONCEPT.39	The following WWT levels shall be transmitted to the TCMS: · WWT level 50% (only provisions) - WWT level 75% (only provisions) - WWT level 100%	Derived Requirement	4S_08.04.-.Catering		N/A
2F_08.03.Zefiro-Europe.CONCEPT.87	The WWT levels shall be sent to the TCMS via dedicated hardwired lines (one for each level).	Derived Requirement	4S_08.04.-.Catering		N/A
2F_08.03.Zefiro-Europe.CONCEPT.88	The WWT levels shall be transmitted to the TCMS via dedicated hardwired lines (one for each level).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_08.03.Zefiro-Europe.CONCEPT.89	The TCMS shall receive the actual level of the WWT and show them on a suitable IDU page (only provisions).	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_08.03.Zefiro-Europe.CONCEPT.90	TCMS shall store an event (Warning) when the WWT is completely full (WWT level 100%).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.3 Function Heating of Water Pipes

3.3.1 Function Design

Heating of water pipes is not needed due to their routing (inside car).

3.3.2 Requirements

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3.4 Function Heating of waste water tanks

3.4.1 Function Design

Waste water tank heating is handled by the Bistrot system (TCMS is not involved). Suitable heating elements fitted on the bottom side of the waste water tank are controlled by a thermostat that activates them when a temperature lower than a preset value (tbd) is detected.
The heating elements will be fed by the medium voltage system (400/230Vac).

3.4.1.1 Normal condition

Suitable heating elements are controlled by a thermostat according to the current temperature of the waste water inside the WWT.

3.4.1.2 Failure cases

a) Heating element/thermostat failure. Loss of WWT heating function.

3.4.1.3 Failure detection

Failure condition detectable during the WWT empting operation (ice forming).

3.4.1.4 Failure handling

a) No action for the driver/crew.Heating element/thermostat has to be changed during maintenance.

3.4.2 Requirements

ID	2F_08.03.Zefiro-Europe.TRS.20
Description	Heating of waste water tanks in the bistro area shall be done to prevent pipe damage.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.03.Zefiro-Europe.CONCEPT.51	In order to prevent freezing of waste water, the waste water tank shall be fitted with suitable heating devices.	Derived Requirement	4S_08.04.-.Catering		N/A
2F_08.03.Zefiro-Europe.CONCEPT.103	The heating devices shall be activated when a temperature below a preset value (TBD) inside the waste waster tank is detected.	Derived Requirement	4S_08.04.-.Catering		N/A
2F_08.03.Zefiro-Europe.CONCEPT.104	The heating devices shall be fed by the medium voltage system (400/230 Vac).	Derived Requirement	4S_08.04.-.Catering		N/A

3.5 Function Freeze Drainage

3.5.1 Function Design

The freeze drainage procedure empties the complete piping of the fresh water system including the fresh water tank.
It is activated by a thermostat when a temperature below a preset value (TBD) in the bistrot piping is detected.
The freeze drainage procedure is available also in case of the Bistrot system is not powered (due to the operational mode of the train).
In addition, it is possible to perform the freeze drainage operation acting on a manual ball valve.

3.5.1.1 Normal condition

A thermostat is used to detect a low temperature in the bistrot piping and activate the empting procedure.

3.5.1.2 Failure cases

a) Empting Electro-valve/thermostat failure. Loss of WWT freeze drainage function.

3.5.1.3 Failure detection

a) Failure condition not detectable.

3.5.1.4 Failure handling

a) No action for the driver/crew. Empting Electro-valve/thermostat has to be changed during maintenance.

3.5.2 Requirements

ID	2F_08.03.Zefiro-Europe.TRS.17
Description	Freeze drainage of water pipes in the bistro area shall be done to prevent pipe damage.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.03.Zefiro-Europe.CONCEPT.57	If a temperature below a preset value [TBD] is detected inside the Bistrot piping, the freeze drainage procedure shall take place. The system shall come back to the normal operation when a temperature above a preset value is detecetd [TBD].	Derived Requirement	4S_08.04.-.Catering		N/A
2F_08.03.Zefiro-Europe.CONCEPT.114	When the freeze drainage procedure is activated, both piping and fresh water tank shall be emptied.	Derived Requirement	4S_08.04.-.Catering		N/A
2F_08.03.Zefiro-Europe.CONCEPT.115	For the freeze drainage function a dedicated power supply line (fed by the train battery) shall be foreseen.	Derived Requirement	4S_08.04.-.Catering		N/A

2F_08.03.Zefiro-Europe.CONCEPT.116	For the freeze drainage function a dedicated permanent power supply line (energized also in case the battery contactors are deactivated) shall be foreseen.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_08.03.Zefiro-Europe.CONCEPT.117	It shall be possible to empty both piping and fresh water tank manually acting on a manual valve.	Derived Requirement	4S_08.04.-.Catering		N/A

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_08.03.Zefiro-Europe.CONCEPT.28			

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_08.03.Zefiro-Europe.CONCEPT.29	FWTankLevel 1 (FWT level 100% - completely full) – only provisions	L	2F_08.03.Zefiro-Europe.CONCEPT.27 2F_08.03.Zefiro-Europe.CONCEPT.74
2F_08.03.Zefiro-Europe.CONCEPT.118	FWTankLevel 2 (FWT level 75% full) – only provisions	L	2F_08.03.Zefiro-Europe.CONCEPT.27 2F_08.03.Zefiro-Europe.CONCEPT.74
2F_08.03.Zefiro-Europe.CONCEPT.119	FWTankLevel 3 (FWT level 50% full) – only provisions	L	2F_08.03.Zefiro-Europe.CONCEPT.27 2F_08.03.Zefiro-Europe.CONCEPT.74
2F_08.03.Zefiro-Europe.CONCEPT.120	FWTankLevel 4 (FWT level 25% full) – only provisions	L	2F_08.03.Zefiro-Europe.CONCEPT.27 2F_08.03.Zefiro-Europe.CONCEPT.74
2F_08.03.Zefiro-Europe.CONCEPT.121	FWTankLevel 5 (FWT level 0% - completely empty)	L	2F_08.03.Zefiro-Europe.CONCEPT.27 2F_08.03.Zefiro-Europe.CONCEPT.74
2F_08.03.Zefiro-Europe.CONCEPT.123	WWTankLevel2 (WWT level 50%) – only provisions	L	2F_08.03.Zefiro-Europe.CONCEPT.39 2F_08.03.Zefiro-Europe.CONCEPT.87
2F_08.03.Zefiro-Europe.CONCEPT.132	WWTankLevel2 (WWT level 75%) – only provisions	L	2F_08.03.Zefiro-Europe.CONCEPT.39 2F_08.03.Zefiro-Europe.CONCEPT.87
2F_08.03.Zefiro-Europe.CONCEPT.122	WWTankLevel1 (WWT level 100% - completely Full)	L	2F_08.03.Zefiro-Europe.CONCEPT.39 2F_08.03.Zefiro-Europe.CONCEPT.87

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_08.03.Zefiro-Europe.CONCEPT.30	FWTankLevel 1 (FWT level 100% - completely full) provision	L	2F_08.03.Zefiro-Europe.CONCEPT.27 2F_08.03.Zefiro-Europe.CONCEPT.74
2F_08.03.Zefiro-Europe.CONCEPT.124	FWTankLevel 2 (FWT level 75% full) provision	L	2F_08.03.Zefiro-Europe.CONCEPT.27 2F_08.03.Zefiro-Europe.CONCEPT.74
2F_08.03.Zefiro-Europe.CONCEPT.125	FWTankLevel 3 (FWT level 50% full) provision	L	2F_08.03.Zefiro-Europe.CONCEPT.27 2F_08.03.Zefiro-Europe.CONCEPT.74

2F_08.03.Zefiro-Europe.CONCEPT.126	FWTankLevel 4 (FWT level 25% full) provision	L	2F_08.03.Zefiro-Europe.CONCEPT.27 2F_08.03.Zefiro-Europe.CONCEPT.74
2F_08.03.Zefiro-Europe.CONCEPT.127	FWTankLevel 5 (FWT level 0% - completely empty)	L	2F_08.03.Zefiro-Europe.CONCEPT.27 2F_08.03.Zefiro-Europe.CONCEPT.74
2F_08.03.Zefiro-Europe.CONCEPT.129	WWTankLevel2 (WWT level 50%) provision	L	2F_08.03.Zefiro-Europe.CONCEPT.39 2F_08.03.Zefiro-Europe.CONCEPT.87
2F_08.03.Zefiro-Europe.CONCEPT.133	WWTankLevel2 (WWT level 75%) provision	L	2F_08.03.Zefiro-Europe.CONCEPT.39 2F_08.03.Zefiro-Europe.CONCEPT.87
2F_08.03.Zefiro-Europe.CONCEPT.128	WWTankLevel1 (WWT level 100% - completely Full)	L	2F_08.03.Zefiro-Europe.CONCEPT.39 2F_08.03.Zefiro-Europe.CONCEPT.87

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_08.03.Zefiro-Europe.CONCEPT.31			

4.2 Bus Interface

NOT APPLICABLE: a bus communication between Bistrot system and TCMS is not foreseen.

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_08.03.Zefiro-Europe.CONCEPT.32			

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_08.03.Zefiro-Europe.CONCEPT.33			

5 References

[1] EC01P006858B TRD Bistrot System